

Computer Networks

Lecture 20 Network Layer

Content



- **(CSMA/CD Problems)
pending**
- **Introduction**

Hidden and Exposed Station Problem

In wireless communication networks, multiple stations share the same communication channel.

Two common problems that occur due to signal range limitations are the Hidden Station Problem and the Exposed Station Problem.

Both are related to the MAC layer multiple access protocols such as Carrier Sense Multiple Access with Collision Detection (CSMA/CD) and Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)

Hidden Station Problem

The Hidden Station Problem occurs when two stations cannot hear each other but both can communicate with the same receiver.

Assume three stations: A, B, and C

- A can communicate with B
- C can communicate with B
- A and C cannot hear each other

A ----> B <---- C

(A and C cannot hear each other)

Exposed Station Problem

The Exposed Station Problem occurs when a station unnecessarily stops transmitting because it senses another nearby transmission, even though its transmission would not cause a collision.

Consider four stations: A, B, C, D

- B is transmitting to A
- C wants to transmit to D
- C hears B's transmission and assumes the channel is busy
- But $C \rightarrow D$ transmission would not interfere with $B \rightarrow A$

$A \leftarrow B$ $C \rightarrow D$

(C hears B but could still transmit)

Introduction

The Network Layer is the third layer of the OSI Model. It is responsible for delivering packets from the source host to the destination host across multiple networks.

Its main job is to determine the best path for data to travel and forward packets through routers so that the data reaches the correct destination.

Services Provided by the Network Layer

- Logical Addressing
- Routing
- Packet Forwarding
- Fragmentation and Reassembly
- Congestion Handling
- Inter-networking

Logical Address

For communication across networks, a global addressing scheme is required. The logical address used on the Internet is known as an Internet Protocol address (IP address).

IPv4 Addressing

- Current Internet addressing system uses IPv4.
- IPv4 addresses are 32 bits long.
- This allows a maximum of 2^{32} possible addresses.

IPv6 Addressing

- IPv6 uses 128-bit addresses, providing a much larger address space.
- It offers greater flexibility in address allocation.

IPV4

An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a device to the Internet.

Address Space: It is the total no. of possible addresses in the protocol

If a protocol uses N bits

Total addresses : 2^N

IPv4 uses 32-bit addresses, which means that the address space is

$$2^{32} = 4,294,967,296$$

IPV4 Notation

There are two prevalent notations to show an IPv4 address: binary notation and dotted decimal notation.

Binary Notation

- In binary notation, the IPv4 address is displayed as 32 bits. Each octet is often referred to as a byte.
- The following is an example of an IPv4 address in binary notation:

01110101 10010101 00011101 00000010

Dotted Decimal Notation

- To make the IPv4 address more compact and easier to read, Internet addresses are usually written in decimal form with a decimal point (dot) separating the bytes.
- The following is an example of an IPv4 address in dotted decimal notation:

117.149.29.2

Question

1. Change the following IPv4 addresses from binary notation to dotted-decimal notation.
 - a) 10000001 00001011 00001011 11101111
 - b) 11000001 10000011 00011011 11111111
2. Change the following IPv4 addresses from dotted-decimal notation to binary notation.
 - a) 111.56.45.78
 - b) 221.34.7.82
3. Find the error, if any, in the following IPv4 addresses.
 1. 111.56.045.78
 2. 221.34.7.8.20
 3. 75.45.301.14
 4. 11100010.23.14.67

Classful Addressing

In classful addressing, the address space is divided into five classes: A, B, C, D, and E.

We can find the class of an address when given the address in binary notation or dotted-decimal notation.

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–239			
Class E	240–255			

b. Dotted-decimal notation

Classful Addressing

In classful addressing of an IP address is divided into two parts:

- Network ID (NetID)
- Host ID (HostID)

These parts help identify which network a device belongs to and which specific device (host) within that network.

In class A, one byte defines the netid and three bytes define the hostid. In class B, two bytes define the netid and two bytes define the hostid. In class C, three bytes define the netid and one byte defines the hostid.

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>
A	11111111 00000000 00000000 00000000	255.0.0.0
B	11111111 11111111 00000000 00000000	255.255.0.0
C	11111111 11111111 11111111 00000000	255.255.255.0

Classful Addressing

One problem with classful addressing is that each class is divided into a fixed number of blocks with each block having a fixed size

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Classfull Addressing

Class A: It is identified by 1st bit of net id. If 1st bit of netid is 0 then those IP addresses belong to class A

IP address range is 0 – 127

Min Value: 00000000 is a DHCP client address

Max Value: 01111111 is a loopback address

All bit Zero: Network Address

All Bit one: Broadcast address

So the class A does not use 0 and 127 and possible network addresses are $2^7 - 2$

Also it has a total of $2^{24} - 2$ host addresses

Classfull Addressing

Class B: It is identified by 2nd and 1st bit of net id. If 1st two bits of netid is 10 then those IP addresses belong to class B

IP address range is 128 – 191

Class B has a total of 2^{14} address

All bit Zero: Network Address

All Bit one: Broadcast address

Also it has a total of $2^{16} - 2$ host addresses

Classfull Addressing

Class C: It is identified by starting 3 bits of net id. If these 3 bits of netid are 110 then those IP addresses belong to class C

IP address range is 192 – 223

Class B has a total of 2^{21} address

All bit Zero: Network Address

All Bit one: Broadcast address

Also it has a total of $2^8 - 2$ host addresses

Classfull Addressing

Class D: It is identified by starting 4 bits of net id. If these 4 bits of netid are 1110 then those IP addresses belong to class D

IP address range is 224 – 239

Class E: It is identified by starting 4 bits of net id. If these 4 bits of netid are 1111 then those IP addresses belong to class D

IP address range is 240 – 255

Question

1. If IP address of the system is 63.12.1.1 to which class does it belong to?
2. If IP address of the system is 160.11.120.19 to which class does it belong to?
3. If IP address of the system is 160.125.257.1 to which class does it belong to?
4. If IP address of the system is 24.31.13.16 . Calculate the netid?
5. If IP address of the system is 24.31.13.16 . What will be the address of 1st host?
6. If IP address of the system is 24.31.13.16 . What will be the address of last host?
7. If IP address of the system is 36.11.119.14 . Find Directed Broadcast Address, 1st host and last host?
8. If IP address of the system is 141.119.89.63 . Find Netid, Directed Broadcast Address, 1st host and last host?
9. If IP address of the system is 199.83.44.254 . Find Netid, Directed Broadcast Address, 1st host and last host?